

MATH 1342 – Elementary Statistics – Week 15: ANOVA and Inference for Slope and Correlation

Dec 1 – 3 | 8.1 Analysis of Variance • 9.1 Inference for Slope and Correlation | Fall 2026 • Section 036 • TTh 6:00–7:20 PM (DIL)

Course-level Objectives	Module-level Objectives	Learning Materials	#	Activities	#	Assessments	#
4. Find and use theoretical probabilities from normal, t, chi-squared and F distributions to form confidence intervals and test hypotheses. Apply the 95% rule to normal and to approximately normal distributions. 5. Analyze relationships between two quantitative variables using correlation and linear regression. 12. Demonstrate skills in inference for regression and ANOVA techniques.	1. Identify when ANOVA is appropriate and state H_0 and H_a in symbols and words. (CO12) 2. Explain what SSG and SSE measure and how the ratio of between-group to within-group variability gives evidence against H_0 . (CO12) 3. Complete an ANOVA table (df, MS, F) from given sums of squares, interpret the F statistic and p-value, and check the conditions. (CO4, CO12) 4. State the simple linear model, distinguish the population slope β_1 and correlation ρ from the sample estimates b_1 and r , and interpret the slope in context. (CO5, CO12) 5. Use regression output to test $H_0: \beta_1 = 0$ (equivalently $\rho = 0$) with a t statistic on $n - 2$ degrees of freedom, state the conclusion in context, and explain why the slope and correlation tests give the same p-value. (CO12) 6. Construct and interpret a confidence interval for the slope, interpret R^2 as the proportion of variability explained, and check the conditions with a residual plot. (CO12)	e-Text: 8.1 Analysis of Variance	MO1, MO2, MO3	Participation: ANOVA – cuckoo-egg example: build the ANOVA table from given SS, find F, decide	MO1, MO2, MO3	Online Homework: 8.1 Analysis of Variance (due Sun Dec 6, 11:59 PM)	MO1, MO2, MO3
		Interactive Lecture: 8.1 Analysis of Variance	MO1, MO2, MO3				
		Slides: 8.1 Analysis of Variance	MO1, MO2, MO3	Participation: Inference for Slope and Correlation – SleepStudy: depression score vs. classes missed (test for correlation and slope, CI for slope, R^2); self-quiz: drinks per week vs. classes missed	MO4, MO5, MO6	Online Homework: 9.1 Inference for Slope and Correlation (due Sun Dec 6, 11:59 PM)	MO4, MO5, MO6
		e-Text: 9.1 Inference for Slope and Correlation	MO4, MO5, MO6				
		Interactive Lecture: 9.1 Inference for Slope and Correlation	MO4, MO5, MO6				
		Slides: 9.1 Inference for Slope and Correlation	MO4, MO5, MO6				
		StatKey – Theoretical Distributions (F)	MO3				